

Adarsh Binu

Software Engineering

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EXPERIENCE

Software / AI Intern

2026 – Present

Talamanda AI

Remote

- Developing **Agentic Marketing**, a multi-agent platform that automates market research, campaign planning, content generation, publishing, lead generation, and outreach.
- Built a modular Python backend around a **Marketing Orchestrator**, coordinating reusable research, planning, content, publishing, and analytics services.
- Implemented research and lead-generation pipelines using **SerpAPI** for market intelligence, business discovery, normalization, deduplication, and prospect scoring.
- Contributed across the **Python** backend and **Next.js, React, TypeScript** frontend, integrating campaign workflows with automated Instagram publishing.
- Deployed the application using **Firebase** and collaborated on development using **Git/GitHub** with AI-assisted engineering tools including **Antigravity IDE**

EDUCATION

B.Tech in Computer Science and Engineering

08/2023 – 05/2027

Amrita Vishwa Vidyapeetham

CGPA: 7.68

Higher Secondary Education

2021 – 2023

Loyola School, Trivandrum

85%

Secondary School Education

2020 – 2021

Loyola School, Trivandrum

91%

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, Java, C, Haskell

AI / ML

PyTorch, TensorFlow, Scikit-learn, DenseNet201, CNNs, PCA, SVM, NLP, LLM Integration, Agentic Workflows

Cloud & DevOps

Docker, Docker Compose, Firebase, Azure, Linux, Git, GitHub

Backend & Web

Flask, Node.js, Next.js, React, REST APIs, JWT, RBAC, PostgreSQL, SQL

Agentic & Integration

Model Context Protocol (MCP), NitroStack, SerpAPI, Gemini AI

Security & Blockchain

JWT, RBAC, SHA-256, ECDSA, Solidity, Smart Contracts, Web Application Security

ONLINE COURSES

Google Cloud Computing Foundations – Cloud Computing Fundamentals (Google Cloud)

Date: May 2026

PROJECTS

Brain Tumor MRI Classification – Hybrid CNN-ML Pipeline

02/2026 – 04/2026

TensorFlow, DenseNet201, PCA, SVM

- Built a four-class medical-image classification pipeline using DenseNet201 feature extraction and PCA-based dimensionality reduction.

- Benchmarked SVM, MLP, and Gaussian Naive Bayes classifiers, achieving 95.66% accuracy and evaluating performance using precision, recall, F1-score, and confusion matrices.

NullTrace – Opinion & Feedback Analytics Platform

10/2025 – 02/2026

Flask, PostgreSQL, Docker, Azure

- Engineered a Flask backend with JWT authentication, RBAC-protected REST APIs, input validation, and a normalized PostgreSQL schema.
- Implemented defenses against SQL injection, XSS, and CSRF alongside bcrypt password hashing, secure file validation, and SHA-256 hash-chained audit logging.
- Integrated NLP sentiment analysis with JSONB-backed analytics endpoints, then containerized the application with Docker Compose and deployed it to an Azure Linux VM.

Hash-Farmers – Decentralized Crop Insurance System

09/2025 – 10/2025

Solidity, Python, SHA-256, ECDSA

- Designed a cryptographic claim-validation pipeline using SHA-256 hashing and ECDSA signatures to authenticate off-chain farmer and event data.
- Integrated off-chain validation with Solidity smart contracts using ecrecover and implemented threshold-based multi-report aggregation for decentralized claim verification.

Echo Runtime – MCP-Based Fraud Detection Platform

07/2026

TypeScript, Node.js, MCP, NitroStack

- Built a fraud-analysis and user-management platform exposing backend capabilities as structured Model Context Protocol (MCP) tools for AI-driven interaction.
- Designed a modular MCP Tool Layer → Service Layer → Application Logic architecture and implemented tools using NitroStack's decorator-based TypeScript framework.
- Integrated application services and external APIs, then tested and deployed a functional end-to-end prototype during the MCP to the Moon Hackathon.

HACKATHONS & INDUSTRIAL TRAINING

MCP to the Moon Hackathon – Protocol X, Echo Runtime

07/2026

Built and deployed a functional MCP-based platform within the hackathon timeline; worked across backend architecture, tool integration, testing, deployment, and team-based Git/GitHub development. Skills: TypeScript, Node.js, MCP, NitroStack, APIs, Git.

BSNL Industrial Training

06/2026

RTTC, Thiruvananthapuram

Completed industrial training in telecommunications infrastructure, networking, transmission systems, fiber-optic communication, and service-provider network operations.

Smart India Hackathon 2025 – Hash-Farmers

2025

Developed the cryptographic and blockchain validation pipeline for a decentralized crop-insurance solution as part of a team submission. Skills: Python, Solidity, SHA-256, ECDSA, Smart Contracts, Git.

LEADERSHIP & COMMUNITY

Mukthyodhayam – Community Development Initiative

09/2025 – 10/2025

Contributed to a police-led community outreach program through field surveys, data collection, and awareness sessions focused on child welfare and substance-abuse prevention.

Seva Volunteering – Amritapuri

10/2023

Supported large-scale public-service coordination and event operations.